Python Lab Assignment: List, Tuple, Set

Instructions:

1. **Duration:** 45 minutes
2. **Total Marks:** 45
3. **All Questions are compulsory**
4. Use **comments and validations** in your code
5. Use **proper identifiers (variable names, function names, etc.)**

Question 1: Find Duplicate Elements in a List

Problem Statement:

Write a program to find all duplicate elements in a given list.

Input:

[1, 2, 3, 2, 4, 5, 1]

Output:

[1, 2]

Question 2: List Rotation

Problem Statement:

Rotate a list to the right by k positions.

Input:

List = [10, 20, 30, 40, 50], k = 2

Output:

[40, 50, 10, 20, 30]

Question 3: Remove Common Elements from Two Lists

Problem Statement:

Remove elements from the first list that are present in the second list.

Input:

List1 = [1, 2, 3, 4], List2 = [3, 4, 5]

Output:

[1, 2]

Question 4: Tuple Element Frequency

Problem Statement:

Count the frequency of each element in a tuple.

Input:

(1, 2, 2, 3, 1, 4)

Output:

{1: 2, 2: 2, 3: 1, 4: 1}

Question 5: Convert List to Tuple and Remove Duplicates

Problem Statement:

Convert a list into a tuple after removing duplicate elements.

Input:
[1, 2, 2, 3, 4, 4]
Output:
(1, 2, 3, 4)

Question 6: Find Maximum and Minimum in Tuple
Problem Statement:
Find the maximum and minimum elements in a tuple without using built-in functions.
Input:
(5, 1, 8, 3)
Output:
Max = 8, Min = 1

Question 7: Set Difference and Intersection
Problem Statement:
Find the intersection and difference between two sets.
Input:
Set1 = {1, 2, 3, 4}, Set2 = {3, 4, 5, 6}
Output:
Intersection: {3, 4}
Difference: {1, 2}

Question 8: Check Subset
Problem Statement:
Check whether one set is a subset of another.
Input:
Set1 = {1, 2}, Set2 = {1, 2, 3, 4}
Output:
True

Question 9: Merge and Sort Unique Elements
Problem Statement:
Merge two lists and return a sorted list of unique elements.
Input:
[3, 1, 2], [2, 4, 5]
Output:
[1, 2, 3, 4, 5]

Question 10: Tuple Pair Sum
Problem Statement:
Find all pairs in a tuple whose sum equals a given value.
Input:
Tuple = (1, 2, 3, 4, 5), Sum = 5

Output:
[(1, 4), (2, 3)]

Question 11: Remove Empty Sets from List
Problem Statement:
Remove all empty sets from a list of sets.
Input:
[{1, 2}, set(), {3, 4}, set()]
Output:
[{1, 2}, {3, 4}]

Question 12: Common Elements in Three Lists
Problem Statement:
Find common elements present in three lists.
Input:
[1, 2, 3], [2, 3, 4], [2, 5, 3]
Output:
[2, 3]

Question 13: Flatten Nested List
Problem Statement:
Flatten a nested list (1 level deep).
Input:
[[1, 2], [3, 4], [5]]
Output:
[1, 2, 3, 4, 5]

Question 14: Symmetric Difference of Sets
Problem Statement:
Find elements that are in either of the sets but not in both.
Input:
Set1 = {1, 2, 3}, Set2 = {3, 4, 5}
Output:
{1, 2, 4, 5}

Question 15: Tuple to Dictionary Mapping
Problem Statement:
Convert two tuples into a dictionary where one tuple contains keys and the other contains values.
Input:
Keys = ('a', 'b', 'c'), Values = (1, 2, 3)
Output:
{'a': 1, 'b': 2, 'c': 3}